4 Conditioning

4.1 First repeat an important definition and theorem.

Def. $\Lambda = \{\lambda_x : x \in \mathcal{X}\}$ is kernel from $\mathcal{X}$ to $\mathcal{Y}$ if $x \mapsto \lambda_x B$ is $\mathcal{A}$-meas $\forall B \in \mathcal{B}$

Thm. If $\exists$ countable $(A_i \times B_i)$, $\mathcal{X} \times \mathcal{Y} = \bigcup A_i \times B_i$, $\lambda_x B_i < \infty$ $\forall x \in A_i$ $\forall i$

then $\forall f \in \mathcal{M}^+(\mathcal{A} \otimes \mathcal{B})$ $y \mapsto f(x,y)$ is $\mathcal{B}$-measurable for ~~all x~~ any fixed x

$x \mapsto \lambda_x^y f(x,y)$ is $\mathcal{A}$-measurable $\forall$

$(\mu \otimes \Lambda)(f) := \mu^x(\lambda_x^y f(x,y))$ defines a measure on $\mathcal{A} \otimes \mathcal{B}$

$\square$

4.2 Product Measure

$\Lambda = \{\lambda\}$ $\mu \otimes \lambda$ is measure on $\mathcal{A} \otimes \mathcal{B}$ defined by linear functional

$\mu \otimes \lambda(f) = \mu^x \lambda_x^y f(x,y) = \mu^x \lambda^y f(x,y)$ $f \in \mathcal{M}^+(\mathcal{A} \otimes \mathcal{B})$

Rmk. σ-finiteness of Λ reduces to σ-finiteness of λ on $\mathcal{Y}$

Thm. <u>Tonelli Theorem</u> If μ is σ-finite on $\mathcal{X}$, λ is σ-finite on $\mathcal{Y}$

$\forall f \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y}, \mathcal{A} \otimes \mathcal{B})$

(1) $\forall x$, $y \mapsto f(x,y)$ is $\mathcal{B}$-meas (2) $x \mapsto \lambda^y f(x,y)$ is $\mathcal{A}$-meas

$\forall y$. $x \mapsto f(x,y)$ is $\mathcal{A}$ (2) $y \mapsto \mu^x f(x,y)$ is $\mathcal{B}$-meas

(3) $(\mu \otimes \lambda)(f) = \mu^x \lambda^y f(x,y) = \lambda^y \mu^x (f(x,y))$

? what's product measurable $\mathcal{A} \otimes \mathcal{B}$

proof. (1) (2) from above theorem

(3) $\mu \otimes \lambda(f) = \mu^x \lambda^y f(x,y)$, then $\lambda^y \mu^x f(x,y)$ is another increasing linear functional with monotone convergence

! used to establish integrability of $f \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y})$

If $f \le g \in \mathcal{M}^+(\mathcal{X} \times \mathcal{Y})$, then $\forall y$, $f(x,y) \le g(x,y)$

$\mu_y^x f \le \mu_y^x g(x,y)$ then $\lambda^y \mu^x f \le \lambda^y \mu^x g$

If $f_n \uparrow f$ then $\mu^x f_n(x,y) \uparrow \mu^x f(x,y)$

$\lambda^y \mu^x f_n \uparrow \lambda^y \mu^x f$

? repeat the unique linear functional def of integral

Also $\forall$ $1_{A \times B}$ $\mu^x \lambda^y 1_{A \times B} = \lambda^y \mu^x 1_{A \times B} = \mu A \lambda B$

so by generating class argument, $\mu^x \lambda^y$ and $\lambda^y \mu^x$ defines the same measure

? Repeat Generating class argument

Cor. <u>Fubini's theorem</u>: For σ-finite μ and λ, and $f \in \mathcal{M}(\mathcal{A}\otimes\mathcal{B})$ such that $\mu\otimes\lambda|f| < \infty$,

(1) $y \mapsto f(x,y)$ is $\mathcal{B}$-meas, for each fixed x, similar for $x \mapsto f(x,y)$

(2) $\lambda^y f(x,y)$ is well-defined, and $\lambda^y f(x,y) < \infty$, μ-a.e.
$x \mapsto \lambda^y f(x,y)$ is μ-integrable

(3) $(\mu\otimes\lambda)f = \mu^x\lambda^y f(x,y) = \lambda^y\mu^x f(x,y)$

? If f is meas wrt completion of $\mathcal{A}\otimes\mathcal{B}$, we need to modify (1) to a.e. version

proof. [1] $\mu\otimes\lambda|f| < \infty \Rightarrow \mu\otimes\lambda f^+ < \infty \quad \mu\otimes\lambda f^- < \infty$

Measurability coming from previous theorem applying $f^+ f^-$

[2] Because $\mu^x\lambda^y f^+(x,y) < \infty$, $\lambda^y f^+(x,y) < \infty$ μ-a.s.

$\lambda^y f^- < \infty$ μ-a.s.

so $\lambda^y f := \lambda^y f^+ - \lambda^y f^- < \infty$ μ-a.s., well-defined

[3] $\lambda^y f = \lambda^y f^+ - \lambda^y f^-$ $\quad \mu^x\lambda^y f = \mu^x\lambda^y f^+ - \mu^x\lambda^y f^-$

$= \lambda^y\mu^x f^+ - \lambda^y\mu^x f^-$

$= \lambda^y\mu^x f$

$\square$

prop. $\mu(f^p) < \infty \Rightarrow \mu\{f > y\} = o(y^{-p})$ as $y \to \infty$

rate for maximal ineq.

proof. $\mu(f^p) = \mu^x(m^y(p y^{p-1}\{y : f(x) > y > 0\}))$

$= \iint p y^{p-1} 1\{y : f(x) > y > 0\}\, dy\, dx$

? why is it valid

then $\mu f^p = p\, m^y(y^{p-1}\mu\{f > y > 0\})$

$\square$

prop. Independence and product measure

$\Rightarrow$ $(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathcal{X}, \mathcal{A})$, $\xrightarrow{Y} (\mathcal{Y}, \mathcal{B})$ $\quad$ If X has marginal distribution $\mu = \mathbb{P}X^{-1}$, Y has distri. $\lambda = \mathbb{P}Y^{-1}$

The joint distribution is $\mathbb{P}(X\times Y)^{-1}$, i.e. a probability measure $\mathbb{Q}$ on $\mathcal{A}\otimes\mathcal{B}$, $\mathbb{Q}D = \{\mathbb{P}\{(X,Y)\in D\}\}$

$\forall D = A\times B$ $\quad \mathbb{Q}D = \mathbb{P}\{(X,Y)\in A\times B\}$ if X, Y Ind.

$= \mathbb{P}\{X\in A\}\,\mathbb{P}\{Y\in B\} = \mu A\cdot\lambda B$

Hence $\mathbb{Q} = \mu\otimes\lambda$

$\Leftarrow$ If $\mathbb{Q} = \mu\otimes\lambda$, then apparently X, Y are independent

$\square$

---

(1) Fubini's theorem

(2) $\mu(f > y) = o(y^{-p})$ for $f \in L^p$

(3) Independence $\Leftrightarrow$ decomposition of joint dist. as product of marginals.

Chapter 37 Empirical Process

1. Set up

$\{W_{Tt} : t \le T, T \ge 1\}$ be a triangular array of $\mathcal{W}$-valued rv from a prob. space $(\Omega, \mathcal{A}, P)$. $\mathcal{W} \subset \mathbb{R}^k$, Let $\mathcal{T}$ be a pseudometric space with p-metri ρ

Def. $\mathcal{M} = \{m(\cdot, \tau) : \tau \in \mathcal{T}\}$ be a class of $\mathbb{R}^s$-valued func. on $\mathcal{W}$. An <u>empirical process</u> $v_T(\cdot)$ by

$$v_T(\tau) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} [m(W_t, \tau) - Em(W_t, \tau)] \quad \text{for } \tau \in \mathcal{T}$$

□

Def. $v_T(\cdot) \rightsquigarrow v(\cdot)$ if $E^* f(v_T(\cdot)) \to Ef(v(\cdot))$ $\forall f \in \mathcal{U}(B(\mathcal{T}))$ that's all bounded, uniformly continuous real func. on $B(\mathcal{T})$

◇

2. Sufficient conditions for weak convergence, $B(\mathcal{T})$: $\mathbb{R}^s$-func.

Def. $\{v_T(\cdot) : T \ge 1\}$ is <u>stochastically equicontinuous</u> if $\forall \varepsilon > 0, \eta > 0$, $\exists \delta$ such that

$$\overline{\lim_{T \to \infty}} P\left[\sup_{\rho(\tau_1, \tau_2) < \delta} \| v_T(\tau_1) - v_T(\tau_2) \| > \eta\right] < \varepsilon$$

◇

prop. If (1) $(\mathcal{T}, \rho)$ is totally bounded

(2) $\forall$ finite subsets $(\tau_1, \cdots, \tau_J)$ of $\mathcal{T}$. $(v_T(\tau_1)', \cdots, v_T(\tau_J)')'$ converges in distribution

(3) $\{v_T(\cdot) : T \ge 1\}$ is stochastic equicontinuous

Then $\exists$ measurable $B(\mathcal{T})$-valued stochastic process $v(\cdot)$, having uniform ρ continuous sample path, $v_T \overset{wp1}{\rightsquigarrow} v$

□

3. Equivalent definitions of stoc. equic.

← 补充

1.5 Modes of Convergence

1.5.1 Definitions for $f: (X, \mathcal{B}, \mu) \to \mathbb{C}$

(1) pointwise convergence	$f_n(x) \to f(x) \quad \forall x \in X$
(2) uniform convergence	$\forall \varepsilon > 0, \exists N$, s.t. $\forall n \geq N$, $\lvert f_n(x) - f(x) \rvert \leq \varepsilon \quad \forall x \in X$
(3) pointwise a.e. convergence	$f_n(x) \to f(x) \quad \mu$-a.e. X
(4) uniform convergencen a.e. essentially uniform, L^∞	$\forall \varepsilon > 0, \exists N, \forall n \geq N \quad \lvert f_n(x) - f(x) \rvert \leq \varepsilon \quad \forall \mu$-a.e. $x \in X$
(5) almost uniformly	$\forall \varepsilon > 0, \exists E \in \mathcal{B}, \mu(E) \leq \varepsilon, f_n \rightrightarrows f$ on E^c
(6) L^1 convergence	$\lVert f_n - f \rVert_{L^1} = \int \lvert f_n(x) - f(x) \rvert \, d\mu \to 0$ as $n \to \infty$
(7) convergence in measure	$\forall \varepsilon > 0 \quad \mu\{\lvert f_n(x) - f(x) \rvert > \varepsilon\} \to 0$

1.5.2 Put them in set language

$f: X \to \mathbb{C}$ or $(Y, \text{metric } d)$

(1) pointwise convergence

$\forall x \in X$

$\forall \varepsilon > 0 \ \exists N$, s.t. $\forall n \geq N$, $\lvert f_n(x) - f(x) \rvert < \varepsilon$

(1) $X = \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{\lvert f_n(x) - f(x) \rvert < \frac{1}{m}\}$

(2) $\forall m, \ E_m = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{\lvert f_n(x) - f(x) \rvert < \frac{1}{m}\} = X$

(2) uniform convergence

$\forall \varepsilon > 0 \ \exists N$ s.t. $\forall n \geq N \ \lvert f_n(x) - f(x) \rvert < \varepsilon \quad \forall x \in X$

depends on the set $E \subset X$ you consider

~~(1)~~ $\cancel{X = \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{\sup_{x \in X} \lvert f_n(x) - f(x) \rvert < \frac{1}{m}\}}$

$\cancel{= \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \bigcap_{x \in X} \{\lvert f_n(x) - f(x) \rvert < \frac{1}{m}\}}$

(3) a.e. convergence

$\exists E \in \mathcal{B}$. $\mu(E) \leq 0$, s.t.

$X \setminus E \subset E_m \quad \forall m$

pointwise convergence can be put into an argument about the domain.